COLLAPSE OF THE NIELSEN-TORSION QUOTIENT ON $G_{4,3}$ AND STRICT AC_3 CERTIFICATES IN $F/K^{(3)}$

ARGUS OPEN RESEARCH PROJECT

Abstract. In the investigation of potential counterexamples to the Andrews–Curtis conjecture, we establish two exact group-theoretic non-separation theorems: (1) on the nilpotent quotient $G_{4,3} = F_4/[F_4^3, F_4]$, the realizable AC_3 determinant locus coincides with the full unit locus $\mathcal{D}_{\mathrm{AC3}} = \mathcal{R}_U$, collapsing the component-conjugation torsion invariant for all generating tuples; and (2) in the untruncated third-derived quotient $F/K^{(3)}$, explicit finite strict Andrews–Curtis programs reduce both candidate presentations P_0 and P_2 to the trivial kernel representative.

1. Main Theorems

Theorem 1.1. (1) *Let $G_{4,3} = F_4/[F_4^3, F_4]$. Under the component-conjugation determinant map, every unit class in $\mathcal{R}_U$ is realized by an AC_3-equivalent generating tuple: $\mathcal{D}_{\mathrm{AC3}} = \mathcal{R}_U$.*

(2) *For the two benchmark balanced presentations P_0 and P_2, explicit finite sequences of strict Andrews–Curtis moves map both tuples to the trivial representative in $F/K^{(3)}$, where $K^{(3)}$ is the third term of the derived series of the normal closure K.*

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